

A Three-Part Network Analysis of MyAnimeList

Fahmida Islam, Digital Humanities and Digital Knowledge, 0001191767
Regina Manyara, Digital Humanities and Digital Knowledge, 0001163423
Martina Marchesi, Digital Humanities and Digital Knowledge, 0001170488
Rumana Mehboob, Digital Humanities and Digital Knowledge, 0001177993
Cristina Vercelli, Digital Humanities and Digital Knowledge, 0001190875

1. Introduction.....	1
2. Problem and Motivation.....	2
3. Datasets.....	3
4. Validity and Reliability.....	4
5. Measures and Results.....	5
6. Conclusion.....	14
7. Critique.....	15
8. Bibliography.....	16

1. Introduction

Our research focuses on examining the connection between users and the anime they interact with or enjoy watching. It uses ideas from the fields of social network analysis, data science and graph theory, placing a particular emphasis on the bipartite graph, which is used to show relationships between two different sets of nodes. The approach makes it possible to see patterns and trends in how people interact with media on digital platforms.

Instead of looking at the media only as technical products, it works with it as cultural products that can create personal experiences and emotions pointing to the fact that user engagement of a media does not only rely on practical aspects but also on stories, emotions and how the user feels after interacting with it.

Earlier studies by Sosa et. al. (2024) incorporates methods from Natural Language Processing (NLP) and Computational Linguistics in order to look through how textual data and their respective symbolic meaning, in the synopses of the shows, influences the formation, cohesion and dispersion of different communities of anime watchers. Similar methods have been incorporated before to study literary texts, political discourses and how people communicate in organizations by looking at words that appear together.

As people interacting with anime on almost a daily basis, we had some doubts and critiques on the analysis and creation of a recommendation system focused on the co-occurrence of words in the synopses; we think that the genre tags are also an important node that interconnects anime to their users. The goal of this paper is not to provide inputs on improving or making new recommendation systems, instead we will focus on comparing and understanding the 2020's dataset on MyAnimeList, which is an

online anime recommendation system that is used by anime watchers to keep track of the anime they've watched.

2. Problem and Motivation

Given the introduction to the work process that we constructed for this network analysis research, we will go over the problems and motivations that pushed this study to come to life. Our work is built upon a research done by Sosa et al. (2024) titled *Analysis of Bipartite Networks in Anime Series: Textual Analysis, Topic Clustering, and Modeling*¹ in which textual analysis and NLP metrics and practices come together with network analysis to examine primarily how textual descriptions from MyAnimeList² influence anime recommendations systems and the formation of communities, in the end affecting the influence of a user watching anime X and anime Y concurrently. The key aspects of their research are based on the idea that the study of ratings and tags for community-formation and traditional recommender systems are unreliable when it comes to examining the full picture of how anime fans are influenced to watch other anime series. They instead make the argument that it is textual descriptions and keywords found in the synopses of each anime which greatly condition the fans. We found this idea to be interesting, but in inspecting their workflow and process we discovered some fallacies that drove us to rethink their research and rework their analysis. The inconsistencies of their analysis were even more apparent to us based on our own first-hand experience within the community as active anime watchers, therefore, we wanted to perform an analysis of the MyAnimeList community that could result in being more reflective of reality.

At first glance, a major issue we found when reviewing the analysis of Sosa et al. (2024) was how they took the textual synopses on MyAnimeList as an authoritative description of an anime, thus completely disregarding the community structure of the website. They did not seem to take into consideration that the textual synopses provided by the website are open to the public, meaning that any user has the potential to write or edit any of these texts. Although this is a process that may undergo moderation, since online communities usually have specific users who also act as administrators and moderators, the implicit democratic configuration of the community depends on very subjective usage of language and keywords. While this may be expected when looking at the influence of comments or reviews left by users, the same cannot be said for anime descriptions which Sosa et al. (2024) used as objective measures of user influence. Furthermore, a lot of keywords they eventually extracted from their analysis were more information-related than thematically descriptive and did not seem very relevant to the aspects of user-influence and content-relations they were interested in studying.

Ultimately, textual synopses on their own are not informative enough for a significant statistical representation of how users may be affected and pushed to watch certain anime series from others. This means that the decision-making process is oversimplified by the act of “reading a summary” about an anime when user A is being drawn from anime A to watch anime B, rather this should be taken as a secondary factor when

¹ Sosa, Juan, Urrego-Lopez, Alejandro, Prieto, Cesar (2024). *Analysis of Bipartite Networks in Anime Series: Textual Analysis, Topic Clustering, and Modeling*. arXiv:2411.03333

² <https://myanimelist.net/>

analysing influence trends. In fact, their analysis ended up excluding key factors such as themes or genres, popularity and ratings, that make up relevant points for community-formation of X and Y anime watchers nonetheless. In the scope of this paper, we argue that the latter parameters are actually more relevant when observing clusters of trends of user-anime and anime-anime networks even if they fall into the category of more traditional observation methods for recommendation systems.

Based on these concerns, our objective here is to work with the same dataset used by Sosa et al. (2024) applying different measures and a different approach to establish connections between users' behaviours and anime-anime connections based on the study of genre tags and ratings. Our goal is to achieve results that will more accurately reflect the relationship between how anime are described and categorised, and user behaviours, specifically localised to the MyAnimeList environment.

3. Datasets

Because we designed this analysis to be a redesign of the one performed by Sosa et al. (2024) we chose to work with the same dataset they used in their analysis. As such we did not need to gather the data ourselves. The dataset in question is publicly available on Kaggle ³ provided by Hernan Valdivieso who webscraped all the pages of MyAnimeList from 2020. The dataset is divided into five csv files each relating to anime data, user data or both and since a couple of them are very big, we have used the Row Zero spreadsheet ⁴ to compare and analyze them better.

Before we began constructing the networks for our analysis we first needed to process and understand how the data was structured. Although the full Kaggle dataset is made of five csv files, we only made use of two for this project, the anime_synopsis dataset and the rating_complete dataset. The anime_synopsis dataset was suitable for our purposes because we were initially only interested in the genre tags attached to each anime and did not need all the other metadata available in the full anime dataset. From the anime_synopsis dataset we processed the data further by dropping the synopsis column as we did not need it for the scope of this study. From there we counted the unique genres found across the dataset as well as the number of anime per genre.

³ <https://www.kaggle.com/datasets/hernan4444/anime-recommendation-database-2020/discussion/352326>

⁴ <https://rowzero.com/workbook/3EB354813F6539975FEAA3DD/0>

anime. From this dataset we used a sample of 10000 users, which would maximise sample size while minimizing computing power.

4. Validity and Reliability

Part of the concern we had with the analysis performed by Sosa et al. (2024) was that they did not seem to take into consideration the validity of the data they were using. While MyAnimeList is very useful in providing insight into anime data and descriptions, it cannot be considered representative of the entire anime community as a whole. The anime community is active across various platforms online and offline, with the location also having an influence on how these communities form and behave. Furthermore, the previous study was performed in 2024, meaning the data they used was 4 years old at the time, while it is now almost 6 years old for our own analysis. However, the public availability of the dataset on Kaggle makes it useful for performing an analysis which can act as a snapshot in time of the anime community on MyAnimeList, which can then be expanded to more communities in future work.

When treating the data we also had to take into consideration the size of the file we were working with. In particular, when it came to the user data, we were presented with a dataset of over 100 million rows. For the sake of the analysis we wanted to perform we found it more beneficial to work with a smaller sample of this dataset in order to have the flexibility to apply more measures without having to worry about processing power.

When it came to sampling the difficulty was that the user data of a single user was spread over multiple rows, with each row representing the entry of a single anime into a users list. As such we could not simply sample from the entire dataset but instead needed to make a separate dataset of unique user IDs to sample from that and then filter through the full user dataset based on this sample. To perform the sample we used the Random Choice function available from the Numpy library in Python⁵.

5. Measures and Results

For our analysis we decided to construct three networks from the Kaggle dataset to better understand the relationship between the genre tags, the anime and the users. From the anime_synopsis.csv dataset we constructed a genre co-occurrence network. From there we used a Louvain algorithm to find communities within the genre network which we could then use to see what influence these genre communities would have in the grouping of anime together and then subsequently users.

Genre Co-Occurance Network

The first network we created was a co-occurrence network of the genres in our dataset. Since we were interested in looking at the relationship between anime and their genre description, it was first necessary to understand how genres interact and intersect with each other across the overall anime network and if we could extract any trends or

⁵ <https://numpy.org/doc/stable/reference/random/generated/numpy.random.choice.html>

patterns. Genre nodes were connected based on co-occurrence in an anime meaning that if two genres appeared together in an anime an edge would be added between them. These edges were then weighted based on the number of anime featuring the pair of genre nodes. As most anime have at least 4 genre tags associated with them, we used the Combination function from the Itertools library in Python⁶ to find all the possible genre pairings in each anime.

This resulted in a network of 41 nodes and 726 edges. The number of nodes is one less than the total number of unique genres because the "Unknown" genre was not included in the network as it had no connection to any of the other genres. We thought this was acceptable since the nature of "Unknown" gives no descriptive information about the anime in question, and since there were only 63 anime under this genre tag, removing them did not have any impact on the rest of the population.

Once the network was constructed we immediately noticed some trends, namely that "Action" appeared to be the hub node, with 40 edges connected to it. Further inspection revealed that the strongest of these edges was the one between the "Action" node and "Sci-Fi" node, with 1294, correlating to the number of anime which feature this combination. This may be a result of the historic popularity of mecha anime and the prevalence of cyberpunk anime dating all the way back to the 1980s - two specific genres which would fall into the category of both "Action" and "Sci-Fi".

We continued with the analysis of this network by employing a clustering algorithm to the genres to see if they tended towards any distinct communities which we could then use to categorise the anime and the anime fans. For this we selected two algorithms to compare in order to see which would work best for our data. We firstly tested our data with the Greedy Modularity Algorithm made available with the Networkx library on Python⁷. This algorithm is based on the work of Clauset et al (2004), who developed a method for community recognition which focuses on maximising modularity ie, finding the largest partition between communities. The algorithm is initiated by placing each node in its own community, then through each cycle pairs of communities are brought together, making sure to maintain the largest modularity. This process repeats until the modularity can no longer be increased.⁸ Because our initial inspection of the genre community revealed the genres were very closely connected, using an algorithm that prioritised modularity was a good fit for the data we had.

⁶ <https://docs.python.org/3/library/itertools.html#itertools.combinations>

⁷

https://networkx.org/documentation/stable/reference/algorithms/generated/networkx.algorithms.community.modularity_max_greedy_modularity_communities.html

⁸

https://networkx.org/documentation/stable/reference/algorithms/generated/networkx.algorithms.community.modularity_max_greedy_modularity_communities.html

Greedy Modularity Algorithm		
Number of Communities	4	
Modularity	0.1199	
Community 1	Number of Genres	13
	Genres	Action, Adventure, Fantasy, Game, Magic, Martial Arts, Mecha, Military, Police, Sci-Fi, Shounen, Space, Super Power
Community 2	Number of Genres	13
	Genres	Comedy, Ecchi, Harem, Parody, Romance, School, Seinen, Shoujo, Shoujo Ai, Shounen Ai, Slice of Life, Sports, Yaoi
Community 3	Number of Genres	11
	Genres	Demons, Drama, Historical, Horror, Josei, Mystery, Psychological, Samurai, Supernatural, Thriller, Vampire
Community 4	Number of Genres	4
	Genres	Cars, Dementia, Kids, Music

Table 2. Greedy Algorithm Results

The results of the greedy algorithm reiterated the tight interconnected nature of the genre network, with a very low modularity score of 0.12. This means that the communities found had no clear separation between them. Nonetheless, 4 communities were found, clustered into general genre themes, namely: “Action & Adventure” for community 1, “Romance & Slice of Life” for community 2, “Dark & Psychological” for community 3 and “Niche” for community 4. However, the distribution of genres across the communities was not very ideal, with community 4 only having 4 genres. Additionally, there were genres in one community which would be better in another community. For example, Josei, which tends to feature more romance and drama stories

would be better suited for the “Romance & Slice of Life” community rather than the “Dark & Psychological” community.

Based on these results we decided to test the Louvain algorithm⁹ which offers a more refined approach to community detection. The Louvain algorithm uses a multilevel approach to optimising modularity in community detection. Instead of brute-forcing modularity, like with the Greedy algorithm, it incorporates a pooling step where detected communities are reassigned as super nodes and the community detection is repeated again with these new community nodes. These steps loop until the highest modularity is reached.

Louvain Algorithm		
Number of Communities	4	
Modularity	0.1223	
Community 1	Number of Genres	10
	Genres	Action, Adventure, Martial Arts, Mecha, Military, Police, Sci-Fi, Shounen, Space, Super Power
Community 2	Number of Genres	15
	Genres	Comedy, Drama, Ecchi, Harem, Josei, Parody, Romance, School, Seinen, Shoujo, Shoujo Ai, Shounen Ai, Slice of Life, Sports, Yaoi
Community 3	Number of Genres	9
	Genres	Demons, Historical, Horror, Mystery, Psychological, Samurai, Supernatural, Thriller, Vampire
Community 4	Number of Genres	7

⁹ <https://python-louvain.readthedocs.io/en/latest/>

expected there to be a strong correlation between the anime and genre communities. We created a bipartite network, with anime nodes forming edges with the genre community nodes based on the genres in each anime. For each anime we counted the number of communities associated with

each genre, the anime was then assigned to the community with the highest count. Anime which has an equal distribution of communities associated to their genres were labelled as bridge anime and were assigned to each community it linked to.

First 20 anime assignments:					
Anime Name	Num Genres	Assigned Communities	Confidence	Is Bridge	
Cowboy Bebop	6	Action & Adventure	0.666667	False	
Cowboy Bebop: Tengoku no Tobira	5	Action & Adventure	0.600000	False	
Trigun	6	Action & Adventure	0.666667	False	
Witch Hunter Robin	6	Action & Adventure	0.333333	False	
Bouken Ou Beet	4	Action & Adventure	0.500000	False	
Eyeshield 21	4	Action & Adventure, Romance & Slice of Life	0.500000	True	
Hachimitsu to Clover	5	Romance & Slice of Life	1.000000	False	
Hungry Heart: Wild Striker	4	Romance & Slice of Life	0.750000	False	
Initial D Fourth Stage	5	Romance & Slice of Life	0.600000	False	
Monster	7	Dark & Psychological	0.571429	False	
Naruto	6	Action & Adventure	0.833333	False	
One Piece	7	Action & Adventure	0.571429	False	
Tennis no Ouji-sama	5	Romance & Slice of Life	0.600000	False	
Ring ni Kakero 1	3	Action & Adventure	0.666667	False	
School Rumble	4	Romance & Slice of Life	0.750000	False	
Sunabouzu	6	Action & Adventure	0.666667	False	
Texhnolyze	4	Action & Adventure	0.500000	False	
Trinity Blood	3	Dark & Psychological	0.666667	False	
Yakitate!! Japan	2	Action & Adventure, Romance & Slice of Life	0.500000	True	
Zipang	6	Action & Adventure	0.500000	False	

Figure 3. Sample of Anime - Genre Community Mapping

As seen in the analysis of the genre network, there is no clear distinction between communities, however, by highlighting the bridge anime in this mapping we got a clearer sense of the division of communities as well as the intersection of the communities. Adding this second network between the anime and genre communities, brings the genres back to their original context, since genres do not exist and interact in a vacuum but rather through the stories of anime. Therefore it makes sense that we would get better and clearer distinctions between communities once the anime was added back as the second part of this newly formed bipartite network.

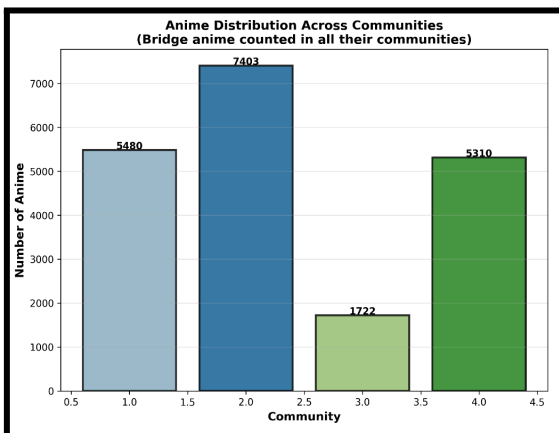


Figure 4.

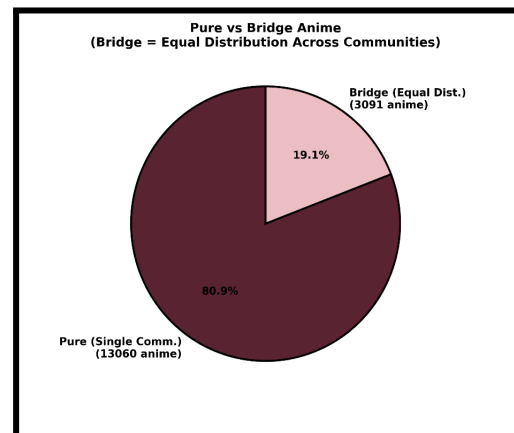


Figure 5.

With the bridge anime identified, we constructed a small genre community matrix, which weighted the connection between communities based on the number of shared bridge anime.

	Action & Adventure	Romance & Slice of Life	Dark & Psychological	Niche
Action & Adventure	0	1023	398	890
Romance & Slice of Life	1023	0	546	1230
Dark & Psychological	398	546	0	393
Niche	890	1230	393	0

Table 4. Genre Community Matrix - weighted with bridge anime

From this matrix we can see strong connections between the “Action & Adventure” and “Romance & Slice of Life” communities based on their shared bridge anime. This makes sense as they are also the communities with the largest anime counts. Interestingly, the strongest connection is between the “Romance & Slice of Life” community and the “Niche” community.

We then applied statistical measures to the larger anime-genre community bipartite network to calculate how well each anime fitted into its assigned communities. Each anime was assigned a confidence score based on how similar one anime was to another anime in its assigned community. The confidence score was measured between 0-1. We found over 50% of the anime had a high confidence score of more than 0.75, validating the genre community assignment as well as the community construction.

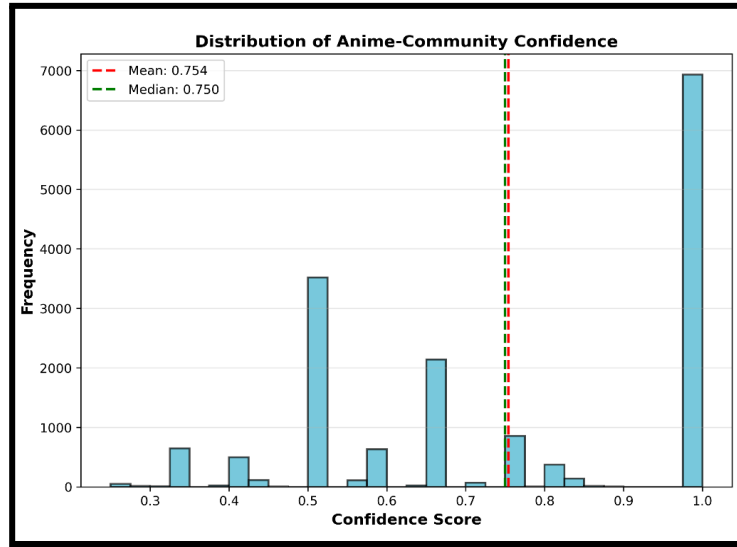


Figure 6. *Anime-Genre Community Confidence Score*

The final network we constructed applied the findings of the other two networks to the user network of MyAnimeList. Since the previous two networks revealed a very strong relationship between genre and anime, it could be assumed that this would then be reflected in the users relationship to the rest of the network. Our initial method for mapping the genre communities to the users was to follow a similar approach to the anime to genre communities i.e counting for each user the number of genre communities associated to the anime in their list and then assigning the user to each community they interacted with. This initial method did not reveal anything insightful as it just resulted in an all to all distribution, where almost all users were assigned to all communities.

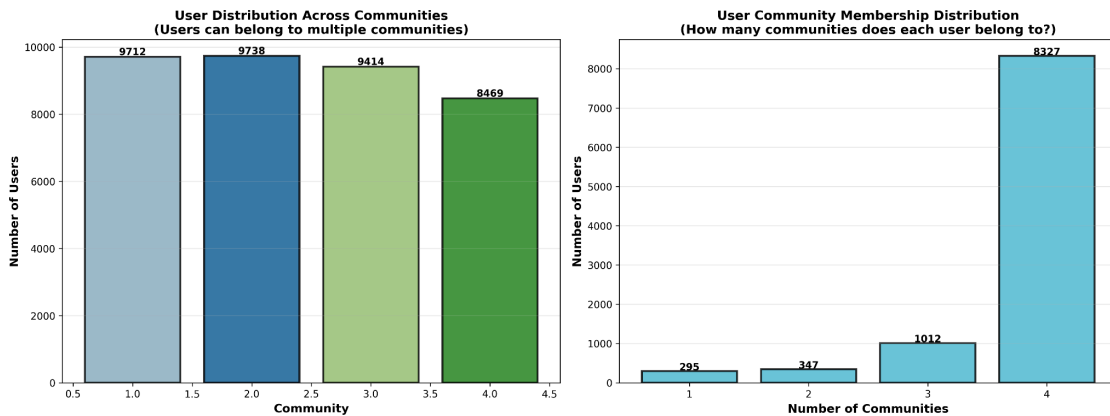


Figure 7. *Results of first attempt of user - genre community network*

While this was not technically incorrect, it makes sense that users would interact with all the genre communities, especially given the generality of the communities, it did not tell us anything about the influence of the genres on the users. Therefore we adjusted our approach to instead consider association to genre communities based on preference

percentage. For each user we looked at all the anime in their list and calculated what percent belonged to each genre community.

Sample user proportion profiles:					
user_id	C1 %	C2 %	C3 %	C4 %	Anime
17	34.4%	45.8%	12.7%	7.1%	704
32	32.7%	42.3%	25.0%	0.0%	44
52	30.5%	53.1%	7.8%	8.6%	216
95	27.8%	50.0%	11.1%	11.1%	17
135	32.9%	41.2%	16.7%	9.2%	306
147	25.5%	57.1%	6.9%	10.4%	449
160	43.8%	30.2%	22.6%	3.5%	243
164	34.5%	56.0%	5.4%	4.2%	148
282	27.3%	63.6%	6.1%	3.0%	61
333	14.0%	67.0%	18.0%	1.0%	93

Figure 8. Sample of user preference percentage table

This shift in focus meant that we could still look at watching trends without having to define strict boxes to fit the users in. A user being more likely to watch action and adventure anime did not negate the fact that they could be watching other anime from other genre communities. With this we were able to not just understand individual user behaviour but translate this back to our genre communities and look at the engagement levels from each community. We also assigned a fifth community to users who had a balanced preference across all the communities.

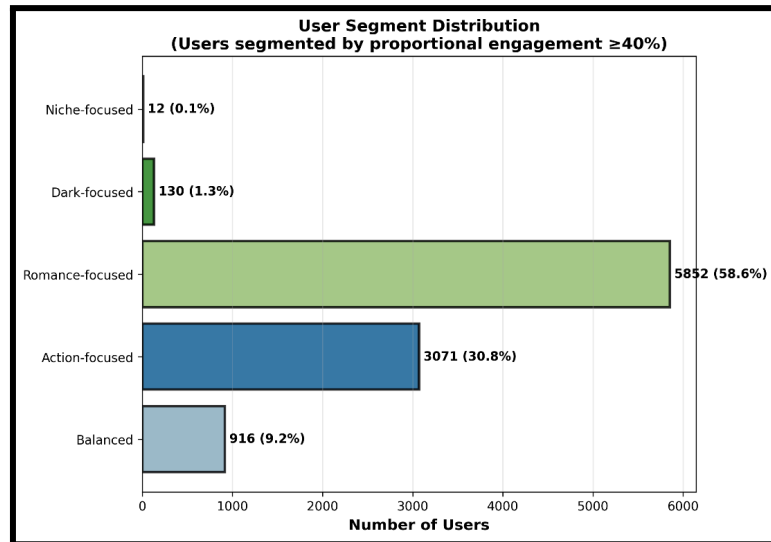


Figure 9.

An immediately interesting result was the majority preference for “Romance & Slice of Life” anime. Considering the genres in this community (Comedy, Drama, Ecchi,

Harem, Josei, Parody, Romance, School, Seinen, Shoujo, Shoujo Ai, Shounen Ai, Slice of Life, Sports, Yaoi) this makes sense as this community offers a wider range of genres and therefore anime, compared to community 1, “Action and Adventure” which was the community with the highest number of anime.

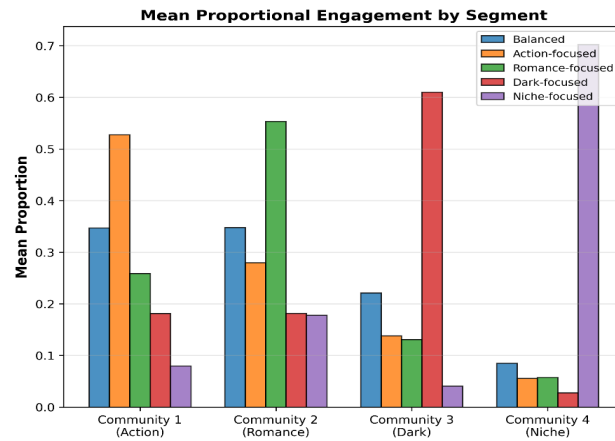


Figure 10.

From the above we can see the trend of how users from each community also interact with other communities. This was important to look at as it highlighted again the interconnected nature of all entities in this anime network, be it genre, anime or anime fans. What we can see for example is that even though “Romance & Slice of Life” had the highest preference percentage, the fans in this community also seemed equally interested in other communities. We can compare this with the “Niche” community which had the highest disparity of fan preference between watching niche anime and watching anything else. We categorised these two behaviours into the “generalist”, a fan who has preference in several communities and the “specialist” a fan who has majority of their preference in only one community. To calculate this distinction we applied a threshold of 40% meaning that if a user had a preference of over 40% they were considered a specialist in that community. The 40% threshold allowed us to identify these unique user behaviours while still preserving the distinction between the communities. From this we can see that although the “Niche” community has the lowest preference rating, it has the highest percent of specialist users meaning that users who watch niche anime tend to stick only to those niche genres.

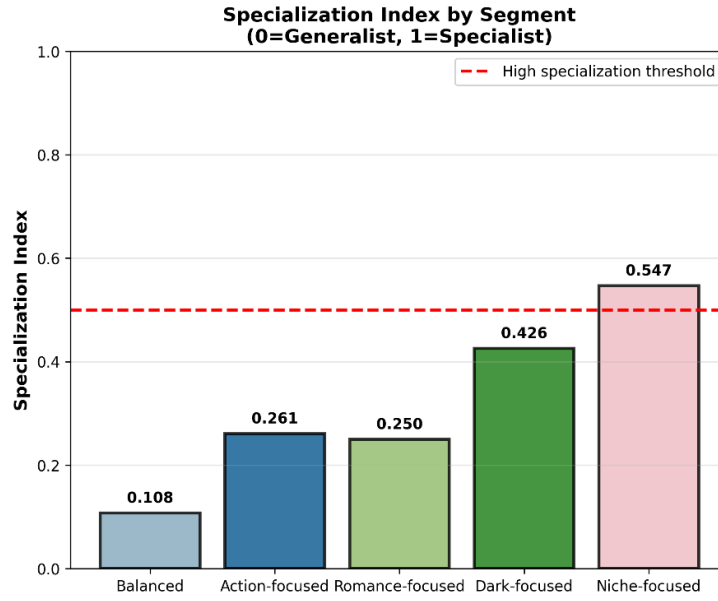


Figure 11.

6. Conclusion

From the construction of the genre co-occurrence network and the anime-genre community bipartite network, we were able to reveal the interconnected nature of the anime community overall. These two networks show that there is a strong relationship between anime and the genres they are described by; one anime cannot be solely taken out and made distinct from the rest but rather they are made to exist in the same space.

Although the modularity scores were low, this reflects the hybrid nature of anime rather than a weakness in the model. The consistent emergence of four communities, such as “Action & Adventure” and “Romance & Slice of Life”, mirrors how well-known series blend genres. For example, *Attack on Titan* combines action, military, and dark psychological elements, while *Your Lie in April* bridges romance, drama, and slice of life. Similarly, fantasy-driven titles like *Sword Art Online* demonstrate how action, game, and romance elements intersect across communities.

The bipartite networks further showed that these genre structures meaningfully shape both anime classification and user behaviour. While many users displayed generalist tendencies across multiple communities, others, particularly those engaging with more niche genres, exhibited specialist patterns. Overall, the findings demonstrate that genres, anime, and users form a tightly connected ecosystem, where boundaries are fluid but still influential in structuring engagement on MyAnimeList.

7. Critique

Reflecting on whether our work has effectively solved the problems identified in the study of Sosa et al. (2024), we identified improvements provided by our approach, but at the same time different limitations remain.

Focusing on genres co-occurrence and user completion habits instead of involving generic terms, as mentioned before, we avoided the contaminated formation of clusters. In addition, we can see genres on MyAnimeList as acting like an influential “language”. Additionally, instead of using arbitrary parameters (like 75% overlap threshold), adopting the Louvian algorithm let communities to arise based on the density of connections; this allowed us to obtain a more stable community structure than those of the reference study. This is a significant point because it shows how smoothly users move between genres without strict pre-defined limits (cutoff).

However some limitations remain in our work, starting from the fact that grounding it on the dataset used in the study of Sosa et al., we worked within the restrictions imposed by it. First of all, relying on a single dataset creates a gap between our model and the complexity of the real world anime community. In addition, looking only at MyAnimeList data, the results obtained cannot be considered representative of the behaviour of anime fans in general; MAL has a specific audience, for the most part they are English-speaking users, settled in the West. Perhaps by looking at data from other platforms such as AniList and Kitty but also Bangumi, Bilibili and Anicri, which are popular in China and Japan, we could analyze how community formation changes.

Another critical issue that remains is that, based on study of Sosa et al., our work can be considered as a snapshot in time: indeed the data dates back to 2020 and in a fast-moving world like the anime production, six years mean a lot in terms of changes. During this period, in fact, the anime industry has undergone several shifts, such as the expansion of certain series that changed genre-crossing habits¹⁰. To overcome this limitation, the study could adopt a longitudinal approach, thus comparing datasets from different years.

Finally, to prove that findings are universal, the model should be tested on other datasets from other platforms, because there could be the risk that the changes made are too focused on the specific features of the Kaggle dataset.

In order to extend this research with the goal of drawing better and more accurate statics about users’ behaviour and anime communities we have rethought the usage of NLP measures and ideas Sosa et al. (2024) used in their study specifically to target comments, possibly also textual synopses, associating them with the parameters of our results. This means to enquire about keywords and relevant thematic concepts associated with the community clusters we found, especially we found to have the strongest connections and most affluence of users and anime-anime nodes.

Another interesting point to consider circles back to how reductive it can be in the digital era we’re living in to regard applying Nature Language Processing to analyse trends ignoring the considerable, if not the majority, of the interactions and influencing happening through platforms of everyday use like social media. For example, conducting research when using the same NLP metrics on comments from social media platforms when sorted through the use of specific tags or filters may be more fruitful to get actually interconnected and more realistic clusters of anime communities. Another aspect to pay attention to is the kind of creative content that can be found online and is shared among the anime community such as fan arts or mixed media like *Anime Music*

¹⁰ <https://aja.gr.jp/english/japan-anime-data>

Videos (AMVs), also known as *fan edits* (edited clips and images framing a specific anime, character, dynamic and more)¹¹. This new variety of fan-made, often non-profital, promotional content¹² is what offers a good insight and potential realistic connection-making between users and different kinds of anime. Examining these types of media might be even more interesting if incorporated with machine learning algorithms as well for images and videos recognition¹³.

8. Bibliography

Clauset, A., Newman, M. E., & Moore, C. "Finding community structure in very large networks." *Physical Review E* 70(6), 2004.

Ito, Mizuko (2010). The rewards of non-commercial production: Distinctions and status in the anime music video scene. *First Monday*, Volume 15, Number 5 - 3. (<https://firstmonday.org/ojs/index.php/fm/article/download/2968/2528>)

Glose, Noah, "Fandom "For You": Understanding the Perceptions, Experiences, and Motivations of TikTok Creators in Anime Fandom" (2024). *West Chester University Master's Theses*. 312. (https://digitalcommons.wcupa.edu/all_theses/312)

Shu, Jing. (2026). Digital memory practices of anime nostalgic fan communities: from the perspective of mediated memory. *International Communication of Chinese Culture*. 10.1007/s40636-026-00355-4.

Sosa, Juan, Urrego-Lopez, Alejandro, Prieto, Cesar (2024). *Analysis of Bipartite Networks in Anime Series: Textual Analysis, Topic Clustering, and Modeling*. arXiv:2411.03333

<https://numpy.org/doc/stable/reference/random/generated/numpy.random.choice.html>

https://networkx.org/documentation/stable/reference/algorithms/generated/networkx.algorithms.community.modularity_max.greedy_modularity_communities.html

<https://docs.python.org/3/library/itertools.html#itertools.combinations>

<https://python-louvain.readthedocs.io/en/latest/>

<https://www.kaggle.com/datasets/hernan4444/anime-recommendation-database-2020/discussion/352326>

<https://myanimelist.net/>

¹¹ Ito, Mizuko (2010). The rewards of non-commercial production: Distinctions and status in the anime music video scene. *First Monday*, Volume 15, Number 5 - 3

¹² Glose, Noah, "Fandom "For You": Understanding the Perceptions, Experiences, and Motivations of TikTok Creators in Anime Fandom" (2024). *West Chester University Master's Theses*. 312.

¹³ Shu, Jing. (2026). Digital memory practices of anime nostalgic fan communities: from the perspective of mediated memory. *International Communication of Chinese Culture*. 10.1007/s40636-026-00355-4.

https://www.researchgate.net/publication/385595738_Analysis_of_Bipartite_Networks_in_Anime_Series_Textual_Analysis_Topic_Clustering_and_Modeling

https://www.canva.com/design/DAG2Kr4uXF0/i_74dQOfczdVwJ7f4Q_7fQ/edit?utm_content=DAG2Kr4uXF0&utm_campaign=designshare&utm_medium=link2&utm_source=sharebutton

<https://rowzero.com/workbook/3EB354813F6539975FEAA3DD/0>